

A Power Model for D-MIMO Systems: Comparing the Energy Consumption of D-MIMO and Collocated MIMO

Thomas Feys, François Rottenberg
KU Leuven, Dept. ESAT-WaveCore, Dramco, B-9000 Ghent, Belgium

Abstract—**TODO**
Index Terms—

I. INTRODUCTION

TODO small paragraph on energy consumption of 5G 6G networks

Recent works have focused on characterizing this energy consumption of wireless networks, and in particular of multiple input multiple output (MIMO) systems. In [1], the authors propose a detailed power consumption model for collocated MIMO systems, which is based on the power consumption of the different hardware components in the system for frequency range 3 (FR3) which is envisioned for 6G systems. Another prevelant enabler for 6G systems is distributed MIMO (D-MIMO)/cell-free MIMO. However, works that characterize the power consumption of D-MIMO systems is limited. **TODO add som refs and work here** Moreover, it is unclear whether the power consumption of D-MIMO systems is on par, lower or higher than the one of collocated MIMO systems. Especially, considering the additional network consumption that occurs from having multiple access point (AP), fronthaul links, synchronization, etc. In this work we introduce an extension of the power model introduced in [1] for the D-MIMO scenario. We incorporate **TODO mention all extensions. TODO mention it is opensource an where it can be found.**

II. D-MIMO SYSTEM MODEL

III. POWER MODEL DESCRIPTION

We first describe the full power consumption of a distributed network (Section III-A), then we focus on the two contributions that are specific for D-MIMO, i.e., the fronthaul and the synchronization of physically separate nodes (Section III-B). Finally we describe the parametric model of the AP consumption, which is based on [1] (Section III-C).

TODO will probably be redundant if we have the sys model Throughout, $l = 1, \dots, L$ indexes the APs, $m = 1, \dots, M$ the antennas of one AP, $k = 1, \dots, K$ the single-antenna users, q the subcarriers, and $i \in \{\text{DL}, \text{UL}\}$ the link direction. The APs are fully digital, so each carries M radio frequency (RF) chains, and the network holds LM antennas in total. The collocated reference is one site of $M_{\text{ant}} = LM$ antennas. Every AP is assumed active and serves every user, and both directions

are evaluated at full physical resource load, $\bar{x}_{\text{DL}} = \bar{x}_{\text{UL}} = 1$, so all APs share one network-wide operating load. **TODO align these symbols with Section II once the system model section is written; ρ_{max} , $P_{T,l}$, R_{DL} , R_{UL} and the coherence block $\tau_c = \tau_p + \tau_u + \tau_d$ are taken from there.**

A. Network Power Consumption

The network consumption collects the L APs, the L fronthaul links that tie them to the central processing unit (CPU), and the CPU itself,

$$P_{\text{net}} = \sum_{l=1}^L \left(P_{\text{AP},l} + \frac{\overline{P_{\text{FH},l}}}{\eta_{\text{FH},\text{sc}}} \right) + \frac{\overline{P_{\text{CPU}}}}{\eta_{\text{CPU},\text{sc}}}, \quad (1)$$

where an overbar denotes a frame-averaged quantity and $\eta_{\text{FH},\text{sc}}, \eta_{\text{CPU},\text{sc}} \in (0, 1]$ are the supply-and-cooling efficiencies of the fronthaul transceivers and of the central unit. The latter acts as an inverse power-usage effectiveness: the CPU sits in a cooled room rather than in an outdoor AP enclosure, so it should not be assigned the values fitted for a macro base station (BS). Each active AP retains the three-block structure of the co-located model from [1] and adds the synchronization of Section III-B2,

$$P_{\text{AP},l} = \frac{\overline{P_{\text{dig},l}}}{\eta_{\text{dig},\text{sc}}} + \frac{\overline{P_{\text{ana},l}}}{\eta_{\text{ana},\text{sc}}} + \frac{\overline{P_{\text{PA},l}}}{\eta_{\text{PA},\text{sc}}} + \frac{P_{\text{sync}}}{\eta_{\text{sync},\text{sc}}}, \quad (2)$$

with the digital, analog and amplifier blocks detailed in Section III-C. The co-located baseline is recovered by setting $L = 1$, $M = M_{\text{ant}}$ and dropping the synchronization, fronthaul and CPU terms.

1) *Functional split*: A single co-located site runs every digital operation in one place. A distributed network does not, and where each operation runs sets both the digital consumption and the fronthaul load. We consider the three splits of Table I. Split S1 forwards precoded samples: the CPU encodes, computes the centralized precoder and applies it, so AP l receives the M streams of frequency-domain samples. Split S2 forwards precoder coefficients and data: the CPU encodes and computes the precoder, but AP l applies it locally. Split S3 is fully local operation: the CPU only encodes, and each AP computes and applies its own precoder from local channel state information (CSI). Splits S1 and S2 realize the same centralized precoder and therefore deliver the same rate; they differ only in where the work runs and in what crosses the fronthaul. S3 delivers a lower rate due to local processing.

TABLE I: Node at which each digital operation runs. The uplink combiner follows the downlink precoder. Channel estimation is listed for completeness but is not priced, since perfect CSI is assumed.

Operation	S1	S2	S3
Encoding / decoding	CPU	CPU	CPU
Channel estimation	CPU	CPU	AP
Precoder computation	CPU	CPU	AP
Precoder application	CPU	AP	AP
OFDM, predistortion, BB filter	AP	AP	AP

2) *MIMO processing and its placement*: **TODO the computational complexities will depend on the precoder type so move this to a table and reference it here.** Following [1], a digital block consumes a static part plus its complex operations per sample Ξ evaluated at the constellation sampling rate $f_{s,I}$ and divided by the computational efficiency η of the implementation. Two MIMO operations must be kept apart, because the split moves them independently. Applying an already computed precoder to K users over N chains costs $2KN$ operations per sample. Computing it costs the Cholesky terms of [1], amortised over the v_{coh} samples of a coherence block. Under centralized operation the CPU inverts the $K \times K$ Gram matrix of the collective channel $\mathbf{H}[q] \in \mathbb{C}^{LM \times K}$, which is the co-located count with the chain number replaced by the total antenna count LM , written here as a network total,

$$\Xi_{\text{pre}}^{\text{cen}} = \frac{1}{v_{\text{coh}}} \left(\frac{K^3}{3} + 3K^2LM + KLM \right). \quad (3)$$

Under local operation AP l instead inverts an $M \times M$ matrix, the same accounting with the antenna and user dimensions exchanged,

$$\Xi_{\text{pre}}^{\text{loc}} = \frac{1}{v_{\text{coh}}} \left(\frac{M^3}{3} + 3M^2K + MK \right). \quad (4)$$

The splits distribute the two counts over the two nodes as

$$\Xi_i^{\text{CPU}} = \iota_{S1} 2KLM + \iota_{S1,S2} \iota_{DL} \Xi_{\text{pre}}^{\text{cen}}, \quad (5)$$

$$\Xi_{i,l}^{\text{AP}} = \iota_{S2,S3} 2KM + \iota_{S3} \iota_{DL} \Xi_{\text{pre}}^{\text{loc}}, \quad (6)$$

where $\iota_{\bullet} \in \{0, 1\}$ equals one for the split or direction named in its subscript. The computation is charged to the downlink only, since time division duplexing (TDD) reciprocity lets the same matrix serve the uplink combiner, while the application is charged in both directions. Every operation appears exactly once across (5), so under S2 the AP is billed only for applying the precoder and the CPU only for computing it. The corresponding power is

$$P_{\text{MIMO}} = \begin{cases} \bar{M}P_s + \frac{f_{s,I}}{\eta} \Xi, & \Xi > 0, \\ 0, & \Xi = 0, \end{cases} \quad (7)$$

with η the computational efficiency of the precoder in the downlink and of the combiner in the uplink, P_s the static power of one FPGA and $\bar{M}$ the number of FPGAs, one per group of 32 chains: $\bar{M} = \lceil M/32 \rceil$ at an AP and $\bar{M} = \lceil LM/32 \rceil$ at the CPU, which processes the collective

array. Charging no static power when $\Xi = 0$ matters for S1, where an AP has no precoder block at all and billing it for an idle FPGA would overstate precisely the split that is meant to be cheap on the AP side.

3) *Central unit*: Channel encoding and decoding stay at the CPU under every split, since that is where the payload enters and leaves the network, and are driven by the delivered network sum rates R_{DL} and R_{UL} through the encoder and decoder models $P_{\text{enc}}, P_{\text{dec}}$ of [1]. With $\mathcal{A}_i$ the frame-averaging operator defined in (16), the central unit consumes

$$\overline{P_{\text{CPU}}} = P_{\text{CPU},0} + \mathcal{A}_{\text{DL}}(P_{\text{enc}} + P_{\text{MIMO,DL}}^{\text{CPU}}) + \mathcal{A}_{\text{UL}}(P_{\text{dec}} + P_{\text{MIMO,UL}}^{\text{CPU}}), \quad (8)$$

with $P_{\text{CPU},0}$ the always-on consumption of the central unit.

B. Fronthaul and Synchronization

1) *Fronthaul*: Following the cell-free power model of [2], the link serving AP l consumes a traffic-independent part covering the optical transceivers at both ends and a part proportional to the rate it carries,

$$P_{\text{FH},l} = P_{\text{FH},0} + \Pi_{\text{FH}} R_{\text{FH},l}, \quad (9)$$

with Π_{FH} in W per bit/s. What $R_{\text{FH},l}$ contains is set by the split of Section III-A1. Let b_{FH} be the number of bits per real sample, so a complex sample costs $2b_{\text{FH}}$ bits. Under S1 the link carries M streams of frequency-domain samples at the constellation rate in both directions and in both the data and the signalling mode,

$$R_{\text{FH},l}^{\text{S1}} = 2b_{\text{FH}} M f_{s,I}. \quad (10)$$

Under S2 the downlink carries the delivered payload plus the precoding coefficients, an $M \times K$ complex matrix per subcarrier refreshed once per coherence block,

$$R_{\text{FH},l}^{\text{S2,DL}} = R_{\text{DL}} + \frac{f_{s,I}}{v_{\text{coh}}} 2b_{\text{FH}} MK, \quad (11)$$

where R_{DL} appears in full because every AP serves every user. Under S3 only the payload is sent over the fronthaul links, $R_{\text{FH},l}^{\text{S3,DL}} = R_{\text{DL}}$. In the uplink, linear combining is distributive over the APs, $\hat{s}_k = \sum_l \mathbf{v}_{kl}^H \mathbf{y}_l$, so an AP holding its combining coefficients forwards K scalar partial sums instead of M sample streams under both S2 and S3,

$$R_{\text{FH},l}^{\text{S2,UL}} = R_{\text{FH},l}^{\text{S3,UL}} = 2b_{\text{FH}} K f_{s,I}. \quad (12)$$

The signalling mode separates the centralized splits from the local one. Centralized operation needs the CPU to see the raw received pilots in order to estimate the collective channel, so during the uplink signalling phase the link carries $2b_{\text{FH}} M f_{s,I}$ under S1 and S2 alike, whereas under S3 the pilots are consumed locally and the signalling load vanishes. Centralization therefore imposes a sample-level fronthaul during pilot transmission independent of the data-phase split.

The link is bidirectional and stays up across the whole frame, so its static term is charged once per link with a single duty cycle rather than once per direction. With

$$\theta_l = \sum_{i \in \{\text{DL,UL}\}} \tau_i [\tau_{i,\text{sig}} + (1 - \tau_{i,\text{sig}}) \bar{x}_i] \quad (13)$$

the fraction of the frame in which the link is in a working mode, and δ_{FH}^μ the reduction factor of an idle transceiver,

$$\overline{P_{\text{FH},l}} = [\theta_l + (1 - \theta_l) \delta_{\text{FH}}^\mu] P_{\text{FH},0} + \Pi_{\text{FH}} \overline{R_{\text{FH},l}}. \quad (14)$$

The averaged fronthaul rate reuses (16) with a zero base level, since an idle link carries no traffic,

$$\overline{R_{\text{FH},l}} = R_{\text{FH},l}^{\text{payload}} + \sum_{i \in \{\text{DL}, \text{UL}\}} \mathcal{A}_i(R_{\text{FH},l}^i, R_{\text{FH},l}^{i,\text{sig}}, 0), \quad (15)$$

where $R_{\text{FH},l}^{\text{payload}}$ is the delivered downlink payload of the data-sharing splits S2 and S3 and is excluded from the averaging, the remaining arguments then being peak rates. The reason is that the map from rate to power in (9) is linear and $\mathcal{A}_i$ with a zero base is linear in its arguments, while the rate model already delivers R_{DL} with its prelog applied. Charging the delivered rate directly is thus equivalent to frame-averaging the corresponding peak rate, and doing both would count the frame structure twice. For the sample-forwarding split S1 the situation is the opposite: (10) is a hardware constant with no dependence on the delivered rate, and the frame structure is then the only mechanism that reduces the fronthaul load.

Remark 1 (Fronthaul ceiling on energy efficiency). *Under a data-sharing split the same payload is duplicated to every AP, so the network fronthaul load is $L R_{\text{DL}}$. Keeping only this term in (1) bounds the downlink energy efficiency by $1/(\Pi_{\text{FH}} L)$, independently of the bandwidth, the transmit power and the precoder. With $L = 16$ and $\Pi_{\text{FH}} = 0.25 \text{ W per Gbit/s}$ the network cannot exceed 250 Mbit J^{-1} however well everything else is designed, which is why the serving cluster size is the dominant lever on cell-free energy efficiency.*

2) *Synchronization*: Coherent joint transmission across physically separate nodes requires the APs to share a frequency and phase reference and requires the TDD chains to be reciprocity-calibrated. Whether the reference is distributed over the fronthaul, over a dedicated timing network, or over the air, the AP must hold it whether or not the frame is carrying data. It is therefore considered as the constant P_{sync} of (2), entirely load-independent and outside the frame averaging, with its own supply-and-cooling efficiency $\eta_{\text{sync,sc}}$.

What P_{sync} covers is the AP-side reference hardware only, that is the disciplined oscillator and the loop that slaves it to a reference. A shared master clock is not a per-AP cost and is counted in $P_{\text{CPU},0}$ instead. The over-the-air part of reciprocity calibration is frame time and CPU computation rather than a continuous per-AP power, and is not modelled. **TODO source a value for P_{sync} and $\eta_{\text{sync,sc}}$. The plausible range for the reference hardware spans roughly two orders of magnitude, from a temperature-controlled oscillator slaved over the fronthaul to an oven-controlled one with its own GNSS receiver, and at $L = 32$ that range moves P_{net} by about 18 %. Report it as a sensitivity band rather than a point value. Over-the-air calibration also consumes frame time, which is not yet charged to $\tau_{i,\text{sig}}$.**

C. Access Point and Co-located Power Consumption

The three frame-averaged blocks of (2) come from the FR3 BS model of [1], evaluated once per AP. A distributed AP carries few antennas, so hybrid beamforming has nothing to amortise and the array is fully digital, $M_{\text{RF}} = M_{\text{ant}} = M$ with no phase shifters. Substituting this into the analog and amplifier assemblies of [1] reduces them term for term to the per-AP expressions below. The digital block cannot be reused directly from [1] because the encoder and decoder move to the CPU and the MIMO processing terms depend on the functional split.

1) *Frame structure and operating-mode averaging*: A TDD frame of duration T_f carries a downlink and an uplink phase with duration ratios $\tau_{\text{DL}}, \tau_{\text{UL}} \in [0, 1]$ and $\tau_{\text{UL}} = 1 - \tau_{\text{DL}}$. Within direction i the time splits into reference signalling, for a fraction $\tau_i \tau_{i,\text{sig}}$, data transmission, for a fraction $\bar{x}_i \tau_i (1 - \tau_{i,\text{sig}})$, micro-sleep, for $(1 - \bar{x}_i) \tau_i (1 - \tau_{i,\text{sig}})$, and idle, for $(1 - \tau_i)$, where $\bar{x}_i \in [0, 1]$ is the average physical resource load. The frame timing is shared by the whole network: under TDD the APs must switch direction together. In micro-sleep and idle a component drops to a fraction $\delta_c^\mu, \delta_c^0 \in [0, 1]$ of a base level P^0 . For a component c with working power P^a during data and P^s during signalling, the frame-averaged consumption of direction i is

$$\begin{aligned} \mathcal{A}_i^c(P^a, P^s, P^0) &= \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) P^a + \tau_i \tau_{i,\text{sig}} P^s \\ &\quad + (1 - \bar{x}_i) \tau_i (1 - \tau_{i,\text{sig}}) \delta_c^\mu P^0 \\ &\quad + (1 - \tau_i) \delta_c^0 P^0. \end{aligned} \quad (16)$$

For the digital and analog blocks the three levels coincide, $P^a = P^s = P^0 = P$, and we write $\mathcal{A}_i^c(P) \triangleq \mathcal{A}_i^c(P, P, P)$; only the amplifier distinguishes them. Collecting the terms proportional to $\bar{x}_i$ separates each averaged consumption into a load-dependent and a load-independent part, which is what allows the consumption to be attributed to the delivered traffic.

TODO make this consistent so this can be removed The rate model states the frame as a coherence block of τ_c samples split into τ_p pilots, τ_u uplink data and τ_d downlink data, whereas the power model works in frame fractions. Charging the pilots to the uplink signalling phase pins the mapping,

$$\tau_{\text{DL}} = \frac{\tau_d}{\tau_c}, \quad \tau_{\text{DL},\text{sig}} = 0, \quad \tau_{\text{UL},\text{sig}} = \frac{\tau_p}{\tau_p + \tau_u}, \quad (17)$$

so that $\tau_{\text{UL}} \tau_{\text{UL},\text{sig}} = \tau_p / \tau_c$ is the pilot share and $\tau_{\text{UL}} (1 - \tau_{\text{UL},\text{sig}}) = \tau_u / \tau_c$ the uplink data share. The same block length also fixes the precoder amortisation, $v_{\text{coh}} = \tau_c$, so that a single number governs both the pilot overhead of the rate model and the reuse of the precoder in (3) and (4).

2) *Power amplifiers*: The amplifiers transmit only in the downlink. With output power p , the instantaneous consumption of one amplifier is

$$P_{\text{PA}}(p) = \frac{1}{\eta_{\text{PA},\text{max}}} [\xi P_{\text{max}} + (1 - \xi) P_{\text{max}}^{1-\alpha} p^\alpha], \quad (18)$$

with P_{max} the maximum output power at a fixed back-off, $\eta_{\text{PA},\text{max}}$ the maximum efficiency, $\xi \in [0, 1]$ the static-to-dynamic ratio and $\alpha \in [0, 1]$ the efficiency exponent. Each AP

is an independent transmitter with its own budget $\rho_{\max}$, so the single site power of [1] becomes the vector of radiated powers $P_{T,l} = \sum_q \sum_k \mathbb{E}(\|\mathbf{w}_{kl}[q]\|^2) \leq \rho_{\max}$ supplied by the rate model, which the two operation modes fill in differently: local operation meets the constraint with equality at every AP, whereas centralized operation rescales all APs by one common factor and brings only the busiest of them to its budget. Assuming the AP power is shared equally across its antennas, $P_{a,l} = P_{T,l}/M$, and

$$\overline{P_{PA,l}} = M \mathcal{A}_{DL}^{PA}(P_{PA}(P_{a,l}), P_{PA}(\zeta_{\text{sig}} P_{\max}), P_{PA}(0)), \quad (19)$$

with ζ_{sig} the fraction of $P_{\max}$ used during signalling. The equal split is not exact, since the per-antenna powers differ for any non-isotropic precoder, but because $\alpha \in [0, 1]$ makes $P_{PA}(\cdot)$ concave, Jensen's inequality gives $\frac{1}{M} \sum_m P_{PA}(P_{a,lm}) \leq P_{PA}(P_{a,l})$: it is a conservative upper bound, tight when the precoder loads an AP uniformly.

Remark 2 (Amplifier sizing). *The relation between $P_{\max}$ and the budget decides how the static term scales with the deployment and must be stated. We size each amplifier for its share of the budget it is actually given, $P_{\max} = \rho_{\max}/M$ at an AP and $P_{\max} = P_{TX}/(LM)$ at the co-located site. Since the comparison holds the total transmit power fixed, $\rho_{\max} = P_{TX}/L$, the two deployments then carry amplifiers of identical rating and differ only in how many sites they sit on. Substituting (18) into (19) makes the data-mode consumption of an AP independent of M ,*

$$M P_{PA}(P_{a,l}) = \frac{\rho_{\max}}{\eta_{PA,\max}} [\xi + (1 - \xi) u_l^\alpha], \quad u_l = \frac{P_{T,l}}{\rho_{\max}}, \quad (20)$$

so the network static floor is $L\xi\rho_{\max}/\eta_{PA,\max} = \xi P_{TX}/\eta_{PA,\max}$, the same for both deployments at equal total transmit power. The per-AP constraint then costs the network twice under centralized operation: only a fraction $\bar{u} < 1$ of the budget is radiated, while the static floor is paid in full. **TODO** source $\xi, \alpha, \eta_{PA,\max}$ for AP-class amplifiers; the values red are fitted for macro-cell hardware.

3) *Analog front end*: The analog block comprises, per RF chain, the digital-to-analog converters (DACs) in the downlink or analog-to-digital converters (ADCs) in the uplink, RF filters, mixers and the low-noise amplifiers, with the local oscillator shared across the chains of one node. Specialising [1] to a fully digital array without phase shifters,

$$P_{\text{ana},DL,l} = P_{LO} + M(2P_{\text{DAC}} + 2P_{\text{filt},RF} + 2P_{\text{mix}}), \quad (21)$$

$$P_{\text{ana},UL,l} = P_{LO} + M(2P_{\text{ADC}} + 2P_{\text{filt},RF} + 2P_{\text{mix}} + P_{\text{LNA}}), \quad (22)$$

the factor two accounting for the in-phase and quadrature branches, and

$$\overline{P_{\text{ana},l}} = \mathcal{A}_{DL}^{\text{ana}}(P_{\text{ana},DL,l}) + \mathcal{A}_{UL}^{\text{ana}}(P_{\text{ana},UL,l}) + P_{\text{ana},\text{misc}}. \quad (23)$$

This is the co-located analog block term for term, with no distributed addition: the synchronization sits outside it, in (2).

The local oscillator is the first term to change character: a co-located array shares one across all M_{ant} chains, whereas the network pays $L P_{LO}$.

4) *Digital signal processing*: Every AP performs the orthogonal frequency division multiplexing (OFDM) modulation and demodulation, the predistortion of its own amplifiers and the baseband filtering whatever the split, so with the component models P_{IFFT} , P_{DPD} and $P_{\text{filt},BB}$ of [1] evaluated at M chains,

$$P_{\text{dig},DL,l} = P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filt},BB} + P_{\text{MIMO},DL,l}, \quad (24)$$

$$P_{\text{dig},UL,l} = P_{\text{IFFT}} + P_{\text{filt},BB} + P_{\text{MIMO},UL,l}, \quad (25)$$

with $\overline{P_{\text{dig},l}} = \mathcal{A}_{DL}^{\text{dig}}(P_{\text{dig},DL,l}) + \mathcal{A}_{UL}^{\text{dig}}(P_{\text{dig},UL,l})$. The encoder and decoder are absent by construction, which is the structural difference from the single-site model, and only the MIMO term of (7) depends on the split.

Remark 3 (Loss of static-power amortisation). *The precoder, combiner and baseband filter share a static power P_s per FPGA, one per group of 32 chains. A co-located array of M_{ant} antennas amortises that power over 32 chains at a time, whereas a network of L APs with $M < 32$ antennas each pays it L times. At $L = 16$ and $M = 4$ the network runs 16 FPGAs where a co-located array of the same 64 antennas runs 2. Chopping a large array into many small ones leaves the per-chain digital costs untouched but destroys the sharing of the fixed ones, and this is a general feature rather than an artefact of the value 32.*

5) *Co-located baseline*: The reference deployment is the unmodified model of [1] evaluated at a single site: one fully digital array of $M_{\text{ant}} = LM$ antennas radiating the full budget P_{TX} , one shared local oscillator, no fronthaul, no central unit and no synchronization power, with the complete digital chain (encoder, precoder, OFDM, predistortion and baseband filter in the downlink; decoder, combiner, OFDM and baseband filter in the uplink) at that one site. It is evaluated through the same parameter set as the distributed network, so the two deployments cannot silently differ in a hardware constant.

D. Delivered Rate and Energy Efficiency

The delivered sum rates R_{DL} and R_{UL} come from the rate model of Section II with the prelogs τ_d/τ_c and τ_u/τ_c already applied, and the network energy efficiency is the delivered rate per consumed watt,

$$\text{EE}_{\text{net}} = \frac{R_{DL} + R_{UL}}{P_{\text{net}}} \quad [\text{bit/J}], \quad (26)$$

which reduces to the single-site energy efficiency of [1] for the co-located baseline. Because a distributed and a co-located deployment cover the same area with the same hardware count, the comparison is made at equal LM and equal total transmit power.

TODO The encoder and decoder are driven by the *delivered* rates and their consumption is then frame-averaged again in (8), so the frame structure enters twice for those two components. This is inherited from [1] and is deliberately

not done for the fronthaul payload in (15); decide which convention the paper adopts and apply it to both.

TODO Effects not yet captured. Fronthaul quantization is modelled only through its rate, so S1 is charged for its fronthaul load but not debited for the signal-to-interference-plus-noise ratio (SINR) loss the b_{FH} bits per sample cause. Channel estimation is unpriced, since perfect CSI is assumed. AP selection is not modelled: all L APs are always active, so the deep-sleep term of (1) and the serving-cluster reduction of Remark 1 are left out. Finally, the reduction factors, the supply-and-cooling efficiencies and the amplifier parameters are inherited from a macro BS and are the most likely source of systematic error when applied to a small outdoor AP.

IV. NUMERICAL EVALUATION

V. CONCLUSION

TODO future work:

- include channel estimation error in rate calculation
- include channel estimation power consumption in power model
- include pilot contamination in rate calculation
- make some parameters more realistic e.g. fronthaul power consumption, synchronization power consumption, etc.
- investigate the impact of hardware impairments on system performance
- accurately characterize all parameters based on the size of the BS/APs.

REFERENCES

- [1] E. Peschiera, S. Yun, Y. Lee, L. V. der Perre, and F. Rottenberg, “A parametric power model of upper mid-band (fr3) base stations for 6g,” in *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 21 476–21 480.
- [2] H. Q. Ngo, L.-N. Tran, T. Q. Duong, M. Matthaiou, and E. G. Larsson, “On the Total Energy Efficiency of Cell-Free Massive MIMO,” *IEEE Transactions on Green Communications and Networking*, vol. 2, no. 1, pp. 25–39, 2018.